

Second Preliminary Report
ECLift
XtraChallenge 2026
March 1st 2026

1 Project Management and Team Organization

The project was carried out by EC-Lift, a team of 10 members from École Centrale de Lyon, using a collaborative approach to ensure effective coordination and steady progress. The objectives and deliverables were defined at the beginning, and the work was divided into smaller tasks with a general timeline to meet the deadline. Regular discussions were conducted to review progress, exchange ideas, and resolve challenges.

Responsibilities were distributed to ensure balanced contribution, and shared documents were used to maintain organization and consistency. This approach enabled efficient teamwork and ensured the successful completion of the preliminary design phase.

2 Aircraft Design Process

2.1 Propulsion

Our propulsion system uses an Emax GT2826 1090KV brushless motor powered by a 4S LiPo battery, delivering up to 2.1 kgf thrust with 85% efficiency for the mission. It is paired with a 12×6 propeller for stable climb and cruise, and controlled by a 60A ESC with cooling ports. The system is mounted on a CNC-machined aluminum plate with a carbon fiber base to reduce vibration and ensure steady flight.

2.2 Aerodynamics

The aerodynamic design is optimized to achieve efficient performance at low Reynolds numbers. The wing uses the NACA 6415 airfoil at the root to provide high lift and the Eppler 66 airfoil at the tip to improve stall behaviour and lift distribution along the span-wise direction. The aircraft also employs a V-tail configuration using the NACA 0012 airfoil, which provides effective longitudinal and directional control with good aerodynamic performance. The overall design minimizes drag and ensures stable and efficient flight.

2.3 Structures

The aircraft structure is being developed with a highly modular design to enable easy assembly, maintenance, and component replacement through simplified connections. The fuselage features an oval cross-section with a slightly squared central section to maximize internal volume, and it is designed to be taller than it is wide to optimize payload accommodation. Electronics are positioned along the longitudinal center of gravity line, slightly aft toward the tail, to maintain stable weight distribution.

3 Aircraft Configuration

The aircraft uses a fixed-wing configuration with a tractor propulsion system for efficient thrust and smooth airflow. A high-wing layout improves stability and flight performance. The wing features a cambered root airfoil and thinner tip airfoil to enhance lift and delay tip stall. A V-tail reduces weight and drag while maintaining control, and the streamlined fuselage minimizes drag and ensures proper integration of onboard components.

4 Materials and Components to be used

The wing and V-tail use CFRP skins with internal balsa ribs and carbon fiber rod spars. The CFRP provides structural strength, the rods carry bending loads, and the balsa ribs maintain the airfoil shape.

The fuselage is built from a carbon fiber composite shell for a lightweight and rigid structure, with internal balsa formers to preserve geometry. A carbon fiber tube connects the fuselage to the V-tail.

The control and telemetry system is powered by an auxiliary battery through a UBEC for reliable power supply. Control surfaces are operated by four servos connected to the receiver and ground controller.

5 Difficulties Encountered

Some challenges were encountered in managing time effectively and ensuring smooth coordination between different design tasks. Maintaining consistency between aerodynamic, structural, and propulsion sections required regular updates and discussions. Additionally, minor integration and compatibility issues arose during the preliminary design stage. These were resolved through team communication, careful review, and iterative improvements.

6 Next Step

Future work will focus on completing the fuselage design, including finalizing the internal structure, component integration, and ensuring proper weight distribution. Further refinement of the aerodynamic model will be carried out to validate performance and incorporate any necessary improvements. Structural analysis will also be conducted to ensure strength and reliability under flight loads. Following the design completion, the project will proceed to fabrication, assembly, and testing to evaluate the overall aircraft performance.

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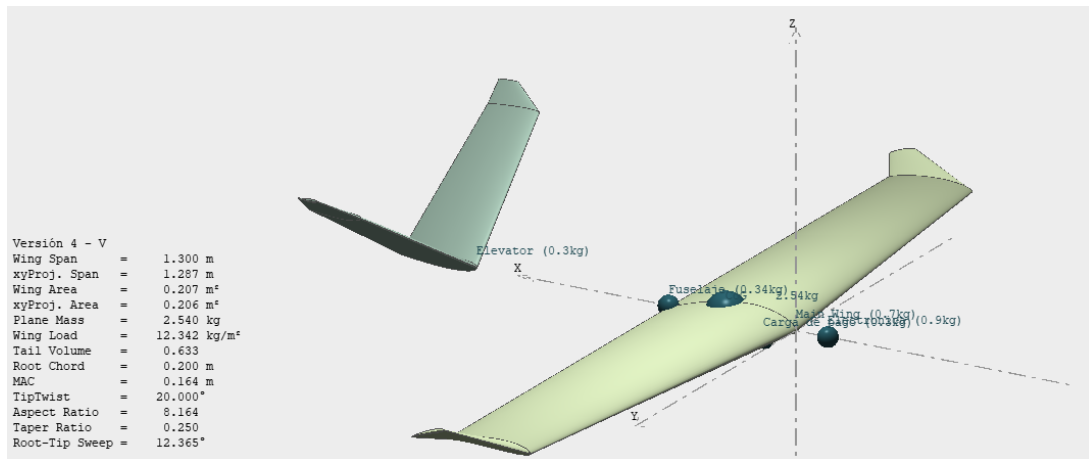
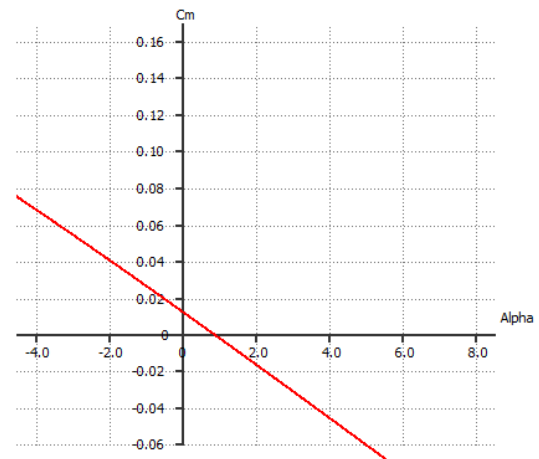


Figure 1: Preliminary configuration

Element	Mass [kg]	x [m]	y [m]	z [m]
Fuselage	0.34	0.2	0	0
Electronics	0.9	-0.05	0	0
Payload	0.3	0.05	0	-0.02
Wing	0.7	0.1	0	0.005
V-tail	0.3	0.563	0	0.064
Global	2.54	0.109	0	0.007

(a) Preliminary mass distribution



(b) Static stability analysis

Figure 2: Mass estimation and longitudinal static stability analysis taking the leading edge at the root as origin of the reference system, x being the longitudinal direction (positive towards the tail), z contained in the symmetry plane (positive towards the top), and y forming a right-handed trihedron

Category	Component	Quantity	Unit Weight (g)	Total Weight (g)
Propulsion	Emax GT2826 Motor	1	175	175
Propulsion	ESC (60A+)	1	90	90
Propulsion	Propeller + Spinner	1	35	35
Propulsion	Motor Mount + Screws	1	60	60
Power	Main Battery 4S 5000mAh	1	350	350
Power	Aux Battery (RX)	1	70	70
Power	XT60 + Power Wiring	1	80	80
Avionics	Telemetry Box	1	150	150
Avionics	Receiver	1	20	20
Avionics	Servos 17g	4	17	68
Avionics	Switch	1	10	10
Avionics	Servo Extensions	6	40	240
Structure	Wing Structure (wood + carbon)	1	400	400
Structure	Fuselage Structure	1	600	600
Structure	Tail Surfaces	1	200	200
Structure	Landing Gear + Wheels	1	350	350
Structure	Linkages + Control Horns	1	150	150
			TOTAL (g):	3048

Figure 3: Components